<HelloWorld />

WEB DEVELOPMENT

Spring 2026

Lecturer: Sanzhar Seitbekov

The Tech Stack

Frontend: Angular

Enterprise Choice. Strict TypeScript.

Backend: Django

"The web framework for perfectionists with deadlines."

The Roadmap

Weeks 1-6
Browser Life (HTML/CSS/JS) → Angular.

Weeks 7-9
Python Dive & Midterm Exam.

Weeks 10-15
Django, APIs, Final Defense.

Goal: Portfolio Project

Grading

Final Exam: 40%

Endterm: 20%

Midterm: 15%

Quizzes/Labs: 25%

⚠ WARNING

Lab Defense is Mandatory.

You must explain your code line-by-line.

AI Policy

At Home

Use ChatGPT as a **Tutor**. Ask for explanations, debug errors.

In Class

Strictly Forbidden on Quizzes & Defenses. If you can't explain it → Ban.

Discipline

Deadlines are Real

Production doesn't wait.

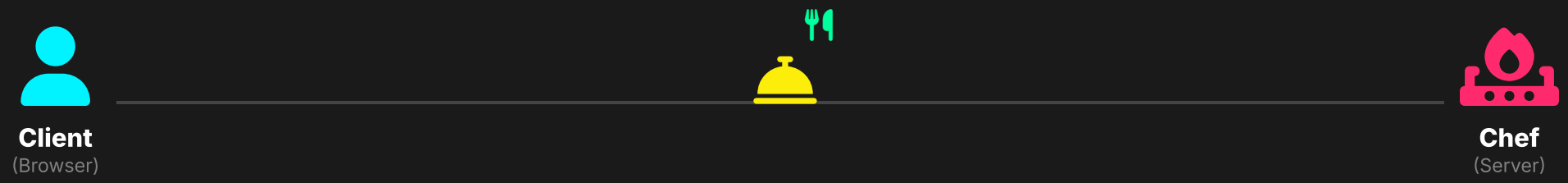
- Attendance > 70% required.
- Miss > 30% = `F (Fail)`

What is the Web?

Not Magic.

It's a global network of machines talking via cables and protocols.

Restaurant Analogy



"Order Steak"
(Request)

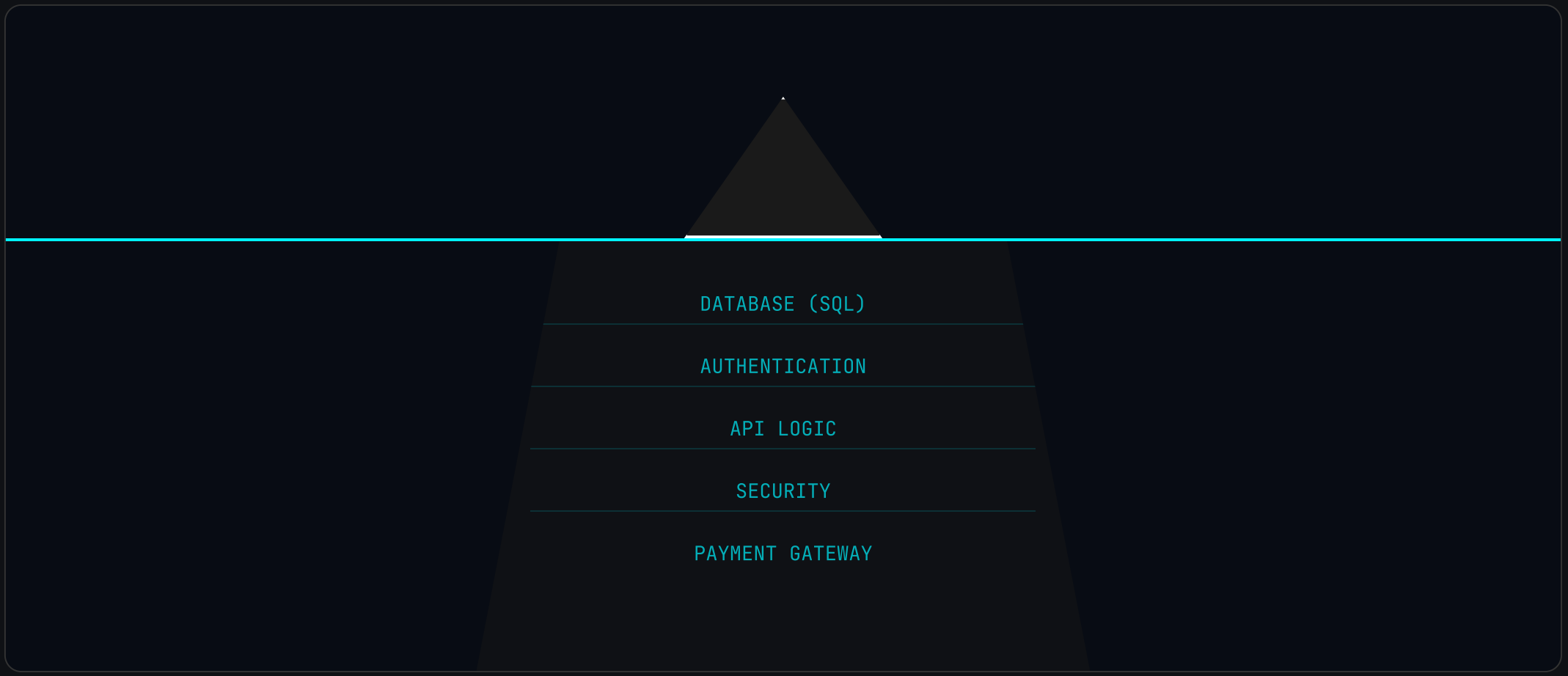
HTTP / Network

"Cook & Serve"
(Response)

Under the Hood

1. **URL:** google.com → **DNS** → 142.250.189.14
2. **Request:** GET /index.html
3. **Server:** Logic + DB Query
4. **Response:** HTML/CSS/JS sent back.

Frontend vs Backend



HTTP Status Codes

200 OK

Success. Here is your data.

201 Created

Success. New record added.

304

Not Modified (Cached).

400 Bad Request

Client Error. Invalid input.

404 Not Found

Resource doesn't exist.

500 Error

Server crashed (Kitchen fire).

10 Minute Break

Relax. We will continue shortly.

Lib vs Framework

Library

You call the code.

"A Toolbox"

Framework

The code calls you.

(Inversion of Control)

Backend Comparison

Tech	Pros	Cons
PHP	Beginner friendly, easy deploy.	Can lack structure (Spaghetti code).
Node.js	Fast, uses JS (Single language).	Not as mature, callback hell.
Java (Spring)	Enterprise standard, compiled.	Verbose, heavy setup.
Python (Django)	"Batteries included", clean syntax.	Need to learn the framework rules.

API

Application Programming Interface

Django builds the API → Angular consumes JSON.

Scaling

Vertical
Bigger Machine
(More RAM/CPU)

Horizontal
More Machines
(Cluster)

Lab #1

Pure HTML & Basic CSS

Structure & Tags.

```
<!— No Frameworks yet —>

<div
class="container"
>


<h1>
My Bio
</h1>


<p>
Web Dev Student
</p>

</div>
```

⚡ Live API Demo

Ask AI

Test the Request/Response cycle.

e.g. Explain 'MVC'...

Response (JSON)

```
// Waiting for input...
```

?

Q & A

Contact & Info

Seitbekov Sanzhar

Lecturer

s.seitbekov@kbtu.kz

Office 178

TEAMS

Join the Team

Team Code: **ksg13tw**